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The growth of catalog sales and the enormous potential of e-commerce elevates the importance of understanding remote purchase. Remote purchase environments differ from traditional bricks-and-mortar purchases in that the purchase decision is more likely to be framed as two separate decisions: consumers' decisions to order and, upon receipt, their decisions to keep or return the item. These two decisions are separated by a period of time, and crucial experiential information often is available only at the second decision point (i.e., after receipt). Consumers' initial lack of experiential information makes product choice more risky. Return policy leniency is one way to minimize the inherent consumer risk, but retailers may avoid instituting overtly lenient policies because they expect increased return rates. However, the endowment effect suggests some surprising benefits of return policy leniency to the retailer. Results from three experiments provide support for the idea that product ownership depends more on perception than possession.

Remote Purchase Environments: The Influence of Return Policy Leniency on Two-Stage Decision Processes

Academic marketing research has begun to focus on the problem of "remote" purchase—as when consumers shop from the home or office rather than on the premises of a bricks-and-mortar store. Interest has been fueled by the growth of e-commerce formats such as the Internet (Alba et al. 1997; Hoffman and Novak 1996), but e-commerce is only part of a general trend toward nonstore shopping that includes catalog shopping and television home shopping.

At present, catalogs dominate this category, with impressive sales figures and growth rates. Mail-order sales have grown steadily at approximately 9.9% a year since 1990 (Niemira 1996). Catalog sales were estimated to exceed $93 billion for 1999 (Miller 1999) and are predicted to be $103 billion by 2001 (Ray and Reis 1996). The Direct Marketing Association predicts that catalogs and e-commerce will account for 6.3% of total retail sales by 2004 (Miller 1999).

How consumers are affected by and adapt to new and existing remote channels is of great interest to both marketing scholars and marketing managers. This article focuses on the behavioral consequences of remote purchases and specifically on the effect of return policies on consumer choice and satisfaction. When consumers shop by catalog or the Internet, their physical remoteness from products creates two key differences from in-store shopping. First, direct examination of the product alternatives is precluded, and therefore the quality of information about sensory product attributes may be inferior (see Alba et al. 1997; Klein 1998; Lynch and Ariely 2000). Second, for nondigital products, customers cannot simply take their purchases home; they must wait for delivery to gather the kind of experiential information that is present during in-store purchases. Both characteristics raise the level of risk to the consumer.

CONCEPTUAL BACKGROUND

The Paradox of Product Returns

Return policies are a source of managerial angst for catalogers and many traditional retailers. Consumers obviously value return policies that are lenient in terms of how long products may be returned after purchase, whether returns are questioned or not, whether cash or store credit is given, and whether sale items can be returned. In contrast, retailers are wary of lenient return policies. Popular managerial theory holds that if a product is easy to return, it is more likely to be returned (Davis, Hagerty, and Gerstner 1998). Not only are more lenient policies costly to operate, but they are also open to consumer abuse.

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Journal of Marketing Research
Vol. XXXVIII (May 2001), 157–169

The lack of a simple solution to this dilemma is evident in the actions of traditional retailers and in the heterogeneity of return policies across catalog companies. Wal-Mart has changed its formerly unrestrictive policies (Ratliff 1994), Talbot's has formed a return policy task force (Cummings 1995), and Target has reported that it has increased the amount of in-store signage devoted to explaining its return policy (Greenberg 1994). Return policy is an especially important issue for remote retailers. Unlike many traditional retailers, the catalog industry has a surprisingly wide variance in return policies (Preddy 1998). Some catalogs do not allow the return of certain items or refund original shipping and handling costs; others not only accept returns unconditionally but also pay for return shipping costs. E-commerce return policies are even more variable, and the onus is on consumers to identify differences in retailers' policies (Italie 1999).

Return Policy Research

Research on the influence of return policy issues is relatively sparse. Hess, Chu, and Gerstner (1996) show that retailers are motivated to impose nonrefundable return charges based on a percentage of the merchandise value to prevent "inappropriate" returns by consumers. Hess and Mayhew (1997) explore a method through which retailers can control returns by identifying either customers or products with high return probabilities. Padmanabhan and Png (1997) investigate how the use of return policies can differentially affect manufacturers' and retailers' profits in markets with certain or uncertain demand. These normative analyses explore ways in which return rates can be reduced or can influence profit but do not examine how such actions affect consumer perceptions or choice processes.

Behavioral investigation of the issue is nonexistent. However, behavioral theory allows predictions about the effects of return policies that are unique to the remote purchase environment. Some key behavioral characteristics differentiate remote sales from more traditional bricks-and-mortar store sales. First, in remote sales, the single purchase is more likely to be framed as two separate decisions: customers' decisions to order and, upon receipt, their decisions to keep or return the item. Second, these two decisions are separated by a period of time. Third, the consumer does not have the opportunity to examine the product physically until the second-stage decision. This research describes how these differences affect choice, commitment, and satisfaction through the influence of both signaling theory and the endowment effect. The findings of this research suggest that consumer behavior in remote purchases is consistent with signaling theory in the initial decision stage (i.e., the order decision). However, subsequent counterintuitive behaviors after ordering represent the more intriguing part of the remote purchase decision process, and the results here suggest that these behaviors can best be explained by the endowment effect.

Decision Conflict, Flexibility, and Endowment

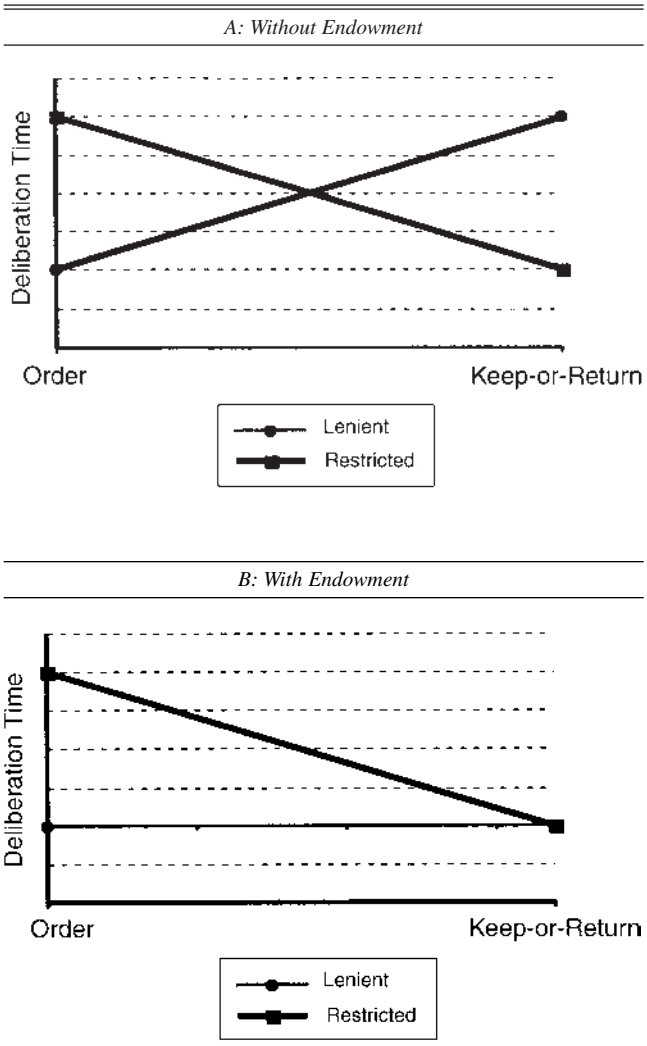
The availability of choice among options inherently involves some level of decision conflict. This conflict is exacerbated by the ability and cost of reversing decisions and the anticipation of regret (Simonson 1992; Tversky and Shafir 1992). A lenient return policy reduces the cost of reversing a bad decision and thus enables consumers to make decisions while maintaining flexibility. If consumers are faced with equivalent alternatives from two catalogs, they will be motivated to choose the catalog with the more lenient return policy. Consumers value flexibility and choose alternatives that allow delay of irreversible decisions (Kahn, Moore, and Glazer 1987). A strict return policy reduces flexibility. Consequently, under conditions of reduced flexibility, consumers should engage in greater deliberation (Janis and Mann 1970; Mann 1971), delay the decision (Tversky and Shafir 1992), or avoid the decision altogether if heightened conflict leads to negative emotion (Luce, Bettman, and Payne 1997).

Lenient policies should increase sales but will increase profits only if return rates do not rise significantly. As previously noted, remote purchases involve two decisions: the decision to order and the decision to keep or return. If consumers have ordered products under lenient return policies with little deliberation, the second-stage decision to keep or return becomes the more important or conflict-laden decision. Therefore, conflict and deliberation at the second stage (i.e., the decision to keep or return) should be high. For costly return policies, the opposite should obtain. Deliberation should be high for the decision to order, because consumers are making a less flexible, more difficult-to-reverse decision. Deliberation should be low for the decision to keep or return, because deep deliberation has already taken place and the costs of return are high. If so, overall purchase deliberation time will be equal for both lenient and restricted return policies but will occur at different decision points, all other things being equal. A stylized illustration of this outcome is presented in Figure 1, Panel A.

However, a different pattern of results is suggested here. Lenient return policies may reduce the total amount of decision conflict (as measured by deliberation time), because consumers fail to compensate for shallow processing during the first stage with more intense deliberation during the second stage (Figure 1, Panel B). The basic rationale is that the endowment effect prompts consumers to view the product in the frame of ownership at the time of product receipt.

The endowment effect (Kahneman, Knetsch, and Thaler 1990) describes a phenomenon in which consumers feel an increase in value for products they own. It has been discussed in terms of a basic self-enhancement mechanism, or "mere ownership" effect (Beggan 1992), and in terms of a more complex loyalty, or "dear ownership" effect (Barone, Shimp, and Sprott 1997). Endowment effects have also been observed at varying levels of ownership duration (Strahilevitz and Loewenstein 1998). Recently, Sen and Johnson (1997) found evidence of endowment or possession effects without actual possession. This finding has interesting implications for remote purchases. Does a consumer truly own a catalog product directly after it is ordered or only after physical receipt? Sen and Johnson's work suggests that even though catalog consumers experience a delay before taking possession of a product, they may behave as owners as soon as they place the order. Thus, the consumers who cavalierly order a product because they know it is easily returned may fail to give sufficient deliberation to the decision to keep or return the product because of a sense of ownership and perceived value.

Figure 1
INFLUENCE OF RETURN POLICY ON PREDICTED DELIBERATION TIME IN A TWO-STAGE REMOTE PURCHASE



Notes: Panels A and B reflect different predictions for deliberation at the keep-or-return decision. Panel B represents an overall reduction in deliberation time for the remote purchase.

H_1: Return policy leniency will decrease deliberation time for the initial decision to order but will not lead to a subsequent increase in the deliberation time for the secondary decision to keep or return.

Thus, lenient return policies may not result in significantly higher return rates. Nonetheless, catalogers may still fear that the added flexibility created by lenient return policies could cause consumers to continue their product search after ordering products or to be more likely to return products after receipt. To understand the validity of this fear, it is important to discuss the potential of lenient return policies to increase commitment to an order relative to restricted return policies.

Quality Signaling and Decision Commitment in Remote Purchases

Businesses send signals about their quality to the consumer (Boulding 1989; Engers 1987; Heil and Robertson 1991; Kihlstrom and Riordan 1984; Nelson 1974; Riley 1975; Simester 1995). Money-back guarantees have been found to be effective as quality signals in retail settings (Moorthy and Srinivasan 1995). The guarantee is perceived as diagnostic, because transaction costs involved in product returns keep low-quality retailers from cheating. The inability of low-quality sellers to profit from false signals is an important tenet of signaling theory. Intuitively, consumers may believe that better return policies are indicative of high-quality catalogers because these firms are standing behind their products. In addition, they may have a naive understanding that return policies are expensive to maintain and may deduce that a retailer would not offer a lenient return policy if a high percentage of products were expected to be returned. Although many bricks-and-mortar retailers have similar return policies, there is a surprising amount of variance in return policy among remote retailers (*Consumer Reports* 1994). Policies differ in terms of both cost (e.g., refund of return costs, processing fees) and time or product restrictions (e.g., a 30-day limit, returns accepted only if packaging is intact).

Thus, even though a lenient return policy may afford the consumer a more flexible and less conflictual order decision, it may also produce expectations of product quality and a stronger commitment to the order. In this case, higher expectations of product quality would be expected under lenient policy conditions. In addition, consumers should exhibit differential prereceipt commitment to the order in behaviors such as continued product search. Strictly on the basis of cost of return, consumers would be expected to search less after ordering goods that have a restricted return policy. However, if a lenient return policy leads to higher expectations of quality, it is plausible that consumers with more flexibility will pursue alternatives less actively during the interval between order and receipt. Therefore, with lenient return policies, the signal of quality may augment commitment in counterforce to the potential decreases in commitment afforded by a liberal return policy.

H_2: Return policy leniency will increase prereceipt expectations of product quality.

H_3: Return policy leniency will reduce continued product search during the period between order and receipt.

Differences Between Remote and Traditional Purchase Environments

This research further posits that the predicted pattern of return policy effects is applicable only to remote purchases and not to more traditional, in-store purchases. In traditional environments, the purchase decision is not as likely to be regarded as a two-stage process involving separate order and keep-or-return decisions. For remote purchases, time may create a perceptual division between the two stages. In remote environments such as catalog or Internet purchases, the order decision is necessarily separated from the keep-or-return decision by a period of time because of product delivery. It has been shown that duration of ownership increases endowment effects (Strahilevitz and Loewenstein 1998). If consumers begin to regard themselves as owning products directly after an order, the delay due to delivery may increase the endowment effects observed at product receipt.

Moreover, consumers lack important experiential attribute information before the order decision in remote pur-

chases. The consumer behavior literature on consumers' inferences from signals shows that consumers rely on such cues primarily when they lack other, more diagnostic bases for judgment (Broniarcyzk and Alba 1994; Rao and Monroe 1989; Simmons and Lynch 1991). In catalog situations, consumers are not able to physically inspect products they order. One plausible hypothesis is that compared with bricks-and-mortar shoppers, remote shoppers might rely more on quality signals such as return policies when ordering. However, when the products are delivered and consumers receive the experiential information that was previously unavailable, perceived differences in quality as a function of return policy should disappear. Analytic signaling models are consistent with this conjecture. Return policies should function as a signal only if they are credible, so that low-quality suppliers cannot cheat by sending high-quality signals, because discrepancies between high-quality signals and low-quality merchandise are obvious to the buyer. This implies that low quality is apparent to the consumer upon direct evaluation.

H_4: Endowment effects in remote purchase are facilitated by (a) the delay between product order and product receipt and (b) the lack of experiential product information at the time of product order.

However, many of the attributes that are unavailable to the consumer during the ordering decision are the same sort of experiential or ambiguous attributes that are difficult to test even with tactile experience. These ambiguous attributes often fall prey to confirmation bias, a phenomenon by which consumers tend to interpret evidence in a manner that confirms their expectations (Deighton 1984; Hoch and Deighton 1989; Hoch and Ha 1986). Thus, it is plausible that the influence of quality signals may persist after receipt, especially when key attributes are ambiguous.

H_5: Differences in perceived quality due to return policy signals can persist after product receipt.

EXPERIMENT 1: ENDOWMENT EFFECTS IN A CATALOG PURCHASE

Before Experiment 1, a preliminary study was conducted in which 77 subjects participated in a conjoint experiment that was designed to test the basic assumption that relatively small differences in return policy attributes can significantly influence catalog choice behavior. Participants were asked to rank similar products (radar detectors) offered by various catalogs. Among other attributes (such as price and radar detection range), two return policy attributes were varied. Results from this study suggest that return policy attributes significantly influence choice probabilities.

Given this evidence of return policy influence, Experiment 1 was designed to test specific hypotheses (H_1, H_2, and H_5) regarding deliberation time and expected quality measures. In this study, participants were offered the opportunity to order a real catalog product.

Method

Twenty-four undergraduates participated in a two-session experiment that was described as a concept-testing study for a proposed new catalog. Sessions contained two to four participants. Return policy was a between-subjects factor with two levels, lenient and restricted. Return policy leniency was manipulated by the ability to return sale items. In the lenient policy condition, sale items could be returned; in the restricted policy condition, sale items could not be returned. Participants first responded to several filler questionnaires and then were shown the catalog. After exposure to the catalog, subjects evaluated several catalog policies in which the return policy, as well as the quality of a subset of the catalog products, was embedded. At the end of the first session, subjects were told that they could order one of the four evaluated catalog products (a candle, a basket, a small penknife, or a barbecue lighter) or take a gift certificate of equivalent dollar value to a local store as a token of appreciation for their participation in the study.[1] They were told that they would receive this item at the second session. The experimenter used a hidden stopwatch to measure the time each subject took to make the decision to order. Participants reconvened two days later for the second session that consisted largely of filler questionnaires. At the end of the second session, subjects were given their ordered products, and the time taken to make the keep-or-return decision (items could be returned for the gift certificate) was recorded. This time measure was again recorded by use of a hidden stopwatch. Finally, participants were asked to evaluate the quality of the product they received.

Results

Deliberation time. The time measures for subjects' duration of decision deliberation (for both the decision to order and the decision to keep or return) were analyzed in a 2 (decision) × 2 (return policy) mixed analysis of variance (ANOVA). Mean times are presented in Table 1. Consistent with H_1, results show that lenient return policies led to reduced deliberation time at the decision to order but did not lead to a subsequent increase in deliberation for the keep-or-return decision. The policy × decision interaction was significant ($F(1,22) = 3.58$, $p < .045$, one-tailed). The time spent deliberating the decision to order was significantly greater for the restricted policy condition (m = 2.31 minutes) than for the lenient policy condition (m = 1.79 minutes; $F(1,22) = 7.24$, $p < .02$). The same is not true in the decision to keep or return, in which deliberation time for the restricted policy condition (m = 1.84 minutes) was similar to that in the lenient policy condition (m = 1.85 minutes; $F(1,22) = .07$).

Quality ratings. A priori, it might be expected that quality evaluations at the time of ordering would be influenced by return policy inferences but that this signaling would not influence evaluations after direct examination at the time of receipt. However, subjects in the lenient policy condition rated products as higher quality both before ordering and after receipt. Table 1 shows that mean product evaluations at the time of ordering were higher for the lenient policy condition (m = 14.1) than for the restricted policy condition (m = 12.5; $F(1,22) = 24.8$, $p < .001$). After the product was received and evaluated, a quality ratings difference remained

[1]Furthermore, a possible limitation exists in that some of the catalog products available to subjects were designated as sale items, and therefore subjects in the restricted policy condition may have believed that these products were not returnable. Although subjects were later told that (because they were participating in market research) they were able to return any product, this may have affected the decision time of some subjects and was thus corrected in Experiments 2 and 3.

Table 1
EXPERIMENT 1 RESULTS

	Quality Rating at Decision to Order	*Quality Rating at Keep-or-Return Decision*	*Deliberation Time at Decision to Order (Minutes)*	*Deliberation Time at Keep-or-Return Decision (Minutes)*
Lenient	14.1	12.4	1.79	1.85
Restricted	12.5	10.3	2.31	1.84
F/t	24.8	6.9	7.24	.07
p-Value	.001	.025	.02	.95
n	24	24	24	24
ω^2	.331	.109	.115[a]	–.019[a]

[a]ω^2 for the policy × decision interaction equals .075.

between the lenient policy condition (m = 12.4) and the restricted policy condition (m = 10.3; F(1,22) = 6.9, $p <$ .025).[2] This supports H_2 and H_5.

Discussion

The results of Experiment 1 demonstrate that return policy influences consumer behavior in catalog purchases. Subjects appear to order products with less consideration when offered a lenient return policy, yet lenient policy subjects did not exercise any more diligence in the keep-or-return decision than did restricted policy subjects. Lenient policies also seemed to signal better quality when subjects judged a product by its photograph in a catalog, yet the quality signal generated by the return policy persisted even after subjects received and directly examined products.

The findings of Experiment 1 suggest that return policy can reduce the overall amount of purchase deliberation and increase quality perceptions. Insofar as ordering a product creates a sense of possession, the quality ratings and final decision times are consistent with prior research on the endowment effect (Sen and Johnson 1997). The results, however, do not speak directly to process. Moreover, Experiment 1 was not designed to examine how the effects of policy leniency vary as a function of delay between ordering and receipt or as a function of the consumer's ability to inspect the product before ordering (H_4). Experiment 2 was designed to address these issues.

EXPERIMENT 2: RETURN POLICY EFFECTS ON REMOTE PURCHASE VERSUS IN-STORE PURCHASE

This experiment includes conditions that mimic the key conceptual differences between remote purchases and in-store purchases. Subjects received full, unambiguous product experience either before the decision to order or after receipt. In addition, subjects experienced either a one-week delay between order and receipt or no delay between order and receipt. This experiment was conducted by computer to increase the precision of decision-time measures.

Selection and Construction of Experimental Stimuli

The cover story used in the main experiment is similar to the one used in Experiment 1. The experimental protocol required subjects to order a catalog product. It was necessary to select a well-known product category with two particular characteristics: unknown brand names and easily manipulated experience. To this end, four Canadian candy bars were chosen. Product experience was manipulated by the opportunity to taste the candy. In the taste-before conditions, subjects tasted the candy bars before ordering. In the taste-after conditions, subjects tasted a small sample of candy after receipt.

Although candy bars may seem like an unusual product category with which to test remote purchase, it is an appropriate choice for several reasons. Candy catalogs do exist. More important, candy bars fit the conceptual constructs being tested, especially the need to manipulate timing of experience through tasting while providing subjects with a real purchase and a believable cover story. The subjects also had an immediate and easy return option that made another better-liked snack available; thus, return of the ordered candy bar was a viable alternative. There may be product attributes (especially credence attributes) for which direct experience does not reduce ambiguity. However, the product category selected for this experiment (candy bars) has been considered to be dominated by experience attributes in prior research (Berger and Mitchell 1989; Wright and Lynch 1995). Note that some bricks-and-mortar stores that sell relatively unfamiliar candy allow in-store tasting (e.g., Ghirardelli, Fannie Mae). The situation in which wrapped candy is sold in self-service stores more closely resembles Experiment 2's remote purchase condition.

Labels from the four selected candy bars were scanned and used to create a mock catalog page. Text was added that detailed nutritional information and price. Another page describing catalog policies was created. Return policy leniency was manipulated through two versions of the policies page: One specified that the vendor would pay for return postage (lenient policy condition), and the other specified that the consumer would be charged $.25 per returned item to cover postage costs (restricted policy condition).

Method

Subjects. Subjects were undergraduates from a large university who received class credit for participation. Subjects participated in groups of two to five. Data were collected from 108 subjects, 11 of whom were eliminated because of computer program problems. Four subjects who failed to attend the second session were eliminated. There was no evidence of differential mortality among conditions. Among

[2]Product evaluations across both conditions decline after product receipt. This is likely, because the quality of the products purchased was mediocre (because of budgetary limitations) and fell short of subjects' expectations. Despite this decline, no subjects chose to return ordered products, because the alternative monetary amount (purchase price) was purposefully set low to encourage order. This product/money trade-off was made more realistic in Experiments 2 and 3; subsequently, more subjects returned products in these studies.

the remaining 93 participants, 85 ordered products and are the respondents for whom data are reported.[3]

Design. A 2 (lenient versus restricted policy) × 2 (immediate versus delayed receipt) × 2 (taste-before versus taste-after experience) × 2 (order versus keep-or-return decision) mixed ANOVA design was used. Return policy, time of receipt, and time of experience were between-subjects factors. Decision was a repeated factor; subjects made both an order and a keep-or-return decision. Subjects were randomly assigned to return policy and time of experience conditions. However, there was a nonrandom assignment to the time of receipt condition. Subjects in immediate conditions were recruited for a one-session experiment, and those in delay conditions were recruited for a two-session experiment. There were two main dependent variables: product quality evaluations and choice deliberation time.

Procedure. The procedure of the main experiment closely followed that of Experiment 1; therefore, only additions to the protocol are discussed here. A visual depiction of the procedure is presented in Figure 2. Subjects were seated at computer carrels and given folders containing a catalog policies page and a product display page. The computer program issued all study instructions and presented the cover story. Subjects were informed that they would be providing consumer information and opinions to a large international candy manufacturer that was considering the use of catalogs to introduce its products in the United States. After responding to several filler questionnaires and evaluating the products and policies featured in the folder, subjects in the taste-before condition were given the opportunity to taste the four featured candies. Next, subjects chose either one of the candies or a gift certificate as a token of appreciation for their participation in the study. Subjects in the taste-after condition did not taste the candies but instead went straight to the product choice task. After the choice task, subjects in the delay condition were reminded to attend the second session and dismissed. Subjects in the immediate condition responded to another questionnaire that tested their memory for product and policy information shown in the mock catalog pages. After the memory task, these subjects were given their chosen product (or gift certificate) and asked to make the keep-or-return decision. At this time, subjects in the taste-after condition were also given a sample of the candy to taste. Finally, subjects evaluated the product they had received and were dismissed. Subjects in the delay condition reconvened one week later and, resuming with the memory questionnaire, followed the same procedure as the subjects in the immediate condition did.

Results

Effect of delay on deliberation time. To determine if the increase in commitment observed in lenient policy catalog purchases is unique to remote purchase scenarios that include a delivery lag, the deliberation times for both the decision to order and the decision to keep or return were compared across the delay and immediate conditions (see Figure 3; means are reported in Table 2). A 2 (lenient versus restricted policy) × 2 (immediate versus delayed time of experience) × 2 (taste-before versus taste-after experience) × 2 (order versus keep-or-return decision) mixed ANOVA design was used, with decision as a repeated factor. For decision deliberation time, there was a significant three-way interaction of time of receipt, return policy, and decision ($F(1,73) = 5.51$, $p = .02$, $\omega^2 = .047$). In the delay condition, there was a significant return policy × decision interaction ($F(1,41) = 5.54$, $p = .02$, $\omega^2 = .075$). Similar to Experiment 1, these results are consistent with Figure 1, Panel B, as hypothesized. In the immediate condition, the two-way interaction was not significant ($F(1,32) = 1.36$, $p = .25$, $\omega^2 = .008$). This pattern of data suggests that return policy affects deliberation time (i.e., H_1 holds) only when there is a delay between order and receipt. When subjects expect an immediate keep-or-return decision, they do not put off the decision-making effort at Time 1. Thus, consistent with H_{4a}, conflict reduction appears to depend on the waiting period caused by delivery.

Comparison of taste-before and taste-after experience. Quality evaluation data also suggest that catalogs differ from bricks-and-mortar stores in the timing of product experience. As suggested in Figure 4, there was a marginally significant planned interaction contrast for the three-way interaction of return policy, time of experience, and decision ($F(1,42) = 3.73$, $p = .06$, two-tailed, $\omega^2 = .049$). For the taste-after condition, the two-way interaction of return policy and decision was significant ($F(1,23) = 13.64$, $p = .02$, two-tailed, $\omega^2 = .275$). Time 1 evaluations were not influenced by return policy ($F(1,27) = 1.92$, $p = .18$). Time 2 evaluations were affected by return policy ($F(1,27) = 5.22$, $p = .03$, two-tailed). Thus, as in Experiment 1, quality expectations appear to influence later experience. However, as predicted, there was no return policy × decision interaction ($F(1,17) = .14$, $p = .72$, $\omega^2 = -.033$) for the taste-before condition. Thus, lenient return policies did not have the same evaluation-based benefits for environments in which consumers received experiential product information before ordering as they did for remote environments in which product experience is delayed until after the initial decision to order. This supports H_{4b}.

Discussion

The results from Experiment 2 suggest that the delay between product choice and receipt and the timing of product experience influence return policy effects. The nonrandom assignment of subjects to the delay versus immediate conditions may be considered a limitation of this study. If so, possible differences in subject motivation due to self-selection into a one-session or two-session study make simple effects of delay difficult to interpret. However, the hypotheses are reflected by observed interactions, and selection cannot explain the full pattern of results demonstrated in the simple interactions.

EXPERIMENT 3: RETURN POLICY EFFECTS ON CONTINUED SEARCH

A third experiment was designed to explore the prediction that with endowment the lenient return policy can engender greater commitment to the order. To this purpose, Experiment 3 investigated both intended order rate and the probability that subjects who had ordered a catalog product would continue product search (H_3) during product delivery delay.

[3]Nonordering subjects did not provide responses for Time 2 measures (keep-or-return decision deliberation time and postreceipt evaluation) in this study. However, initial order decision deliberation time and quality rating data for nonordering subjects are provided in Table 2.

Figure 2
EXPERIMENTAL PROCEDURE

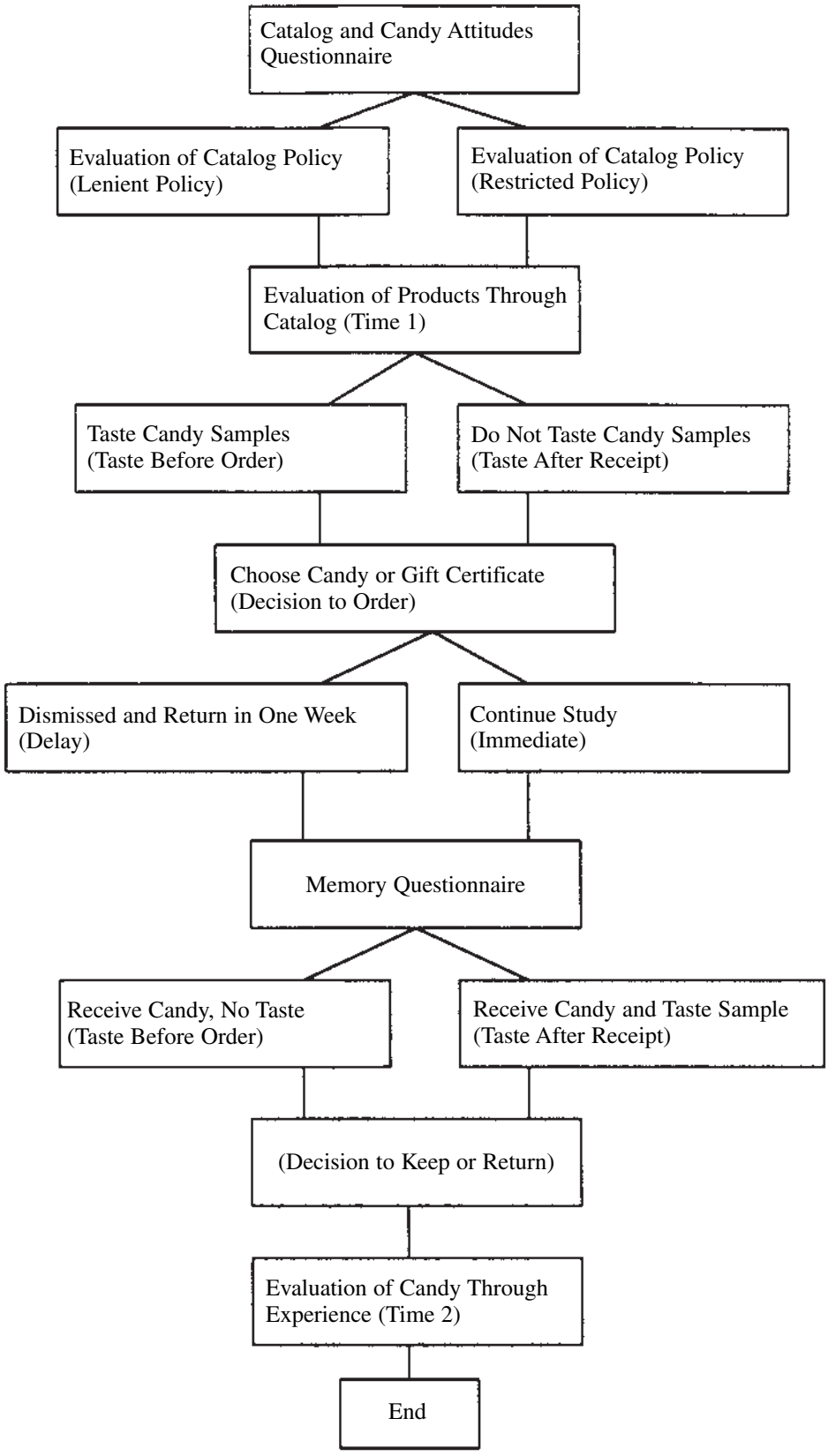
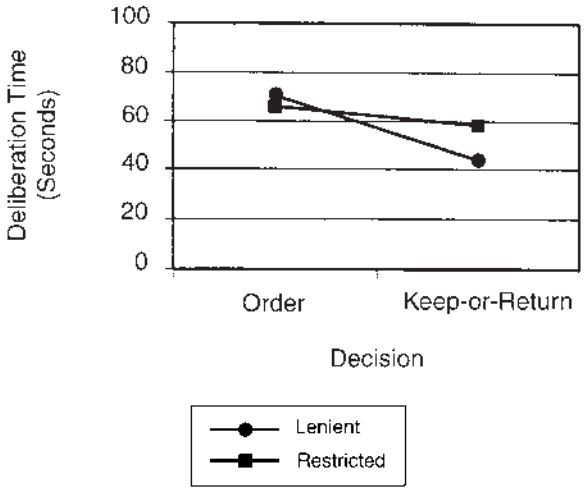
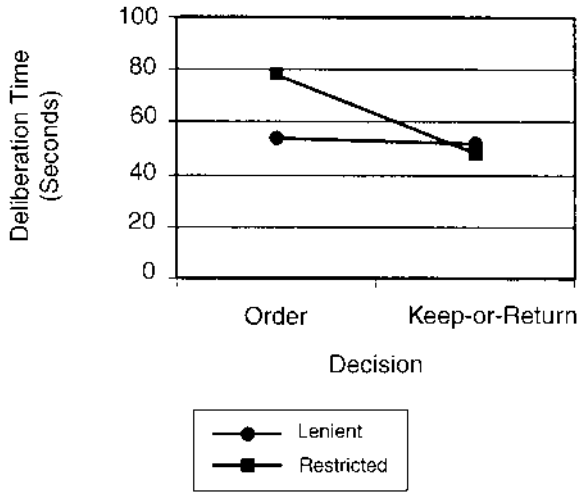


Figure 3
DECISION DELIBERATION TIME: INFLUENCE OF DELAY OF DELIVERY ON COMMITMENT

A: Decision Deliberation Time in Immediate Conditions



B: Decision Deliberation Time in Delay Conditions



Pilot Study 3

An initial pilot study was conducted in which 68 undergraduate participants were presented with one of two catalog purchase scenarios and asked to predict their own behavior. The first scenario asked subjects to imagine that they were considering the purchase of a collegiate T-shirt from a catalog. The catalog's return policy was described as either more or less lenient than similar catalogs. Subjects were asked to predict the probability that they would order the shirt for an upcoming sporting event. Results show that a lenient return policy significantly increased the likelihood of order from 41% to 61% (Table 3). This finding is not surprising, but it illustrates the subject population's rational behavior.

The second scenario was similar to the first but asked subjects to imagine further that they had already ordered the T-shirt from the catalog (whose return policies were again described as more or less lenient than comparable catalogs) but had not yet received it. They were asked to imagine further that they had just heard of a great collegiate clothing store 15 miles from campus. Subjects were asked to predict the likelihood that they would go to this store to continue their search for a T-shirt before the ordered shirt arrived. Results show that even though the restricted return policy would make any shirt chosen at the store more costly because of the subsequent return of the ordered shirt, subjects facing more restricted return policies predicted a greater likelihood of continued search (m = 68%) than did those facing lenient policies (m = 39%; Table 3).

These findings provide preliminary evidence that lenient return rates have utility to consumers and can increase commitment to an order, as evidenced by continued search behavior. It might be expected a priori that people would search less after ordering if it is more costly to reverse the decision to order by returning, but it appears that those ordering under a lenient return policy less actively pursue other alternatives during the interval between order and receipt. Not only is this of managerial interest, but it is also intriguing that consumers might be more likely to continue product search after a catalog order when the subsequent return implied by a successful search would be more costly. Experiment 3 was designed to provide an unobtrusive behavioral measure of continued search as a validation of these self-report findings.

Selection and Construction of Experimental Stimuli

Experiment 3 used the same cover story of catalog concept testing as did Experiments 1 and 2. A new product category, moderately priced pens, was chosen to test for robustness. Another modification to improve the experimental realism of the study involved the dollar amount of the gift certificate offered in exchange for the opportunity to order a product from the catalog. The amount of the gift certificate participants could choose in place of the catalog product (i.e., the "price" of the catalog product) was commensurate with the true value of the item. Therefore, the purchase decision may have been perceived as more real by study participants. As evidence of this, a larger proportion of participants did not order a catalog product compared with Experiments 1 and 2.

A catalog page illustrating three pens was created. The brand names were removed from the images of the pens; participants were told that the unnamed manufacturer of these pens was interested in introducing its own catalog. These catalog pages were placed in hardcover ring binders. A separate booklet contained the other stimulus material, including the description of the catalog policies (embedded in which was the return policy), other filler evaluation measures, and the order form. Another booklet was created for the second session, in which participants were given the choice to keep or return the pen. In addition, a manila folder containing several pages of product information on several competitor's pens was prepared.

Return policy leniency was manipulated through two versions of the return policy page. The lenient policy condition specified that the vendor would refund the original shipping

Table 2
EXPERIMENT 2: MEANS FOR REPORTED INTERACTIONS

	Quality Evaluation Results (Nine-Point Likert Scale)		*Decision Deliberation Time (Seconds)*	
Condition	*Time 1 (Before Order)*	*Time 2 (After Receipt)*	*Decision to Order*	*Keep-or-Return Decision*
Lenient × immediate	n.a.	n.a.	70	44
Restricted × immediate	n.a.	n.a.	66	58
Lenient × delay	n.a.	n.a.	54	52
Restricted × delay	n.a.	n.a.	78	48
Lenient × catalog	3.3	4.0	n.a.	n.a.
Restricted × catalog	3.65	2.1	n.a.	n.a.
Lenient × bricks-and-mortar	2.9	2.7	n.a.	n.a.
Restricted × bricks-and-mortar	3.2	3.2	n.a.	n.a.
Nonordering subjects		n.a.	n.a.	
Lenient policy	3.22		60	
Restricted policy	3.63		51	

Notes: n.a. = not applicable.

and handling cost plus return postage, a value of $1.50; the restricted policy condition specified that the amount would not be refunded. This study also attempted to replicate Experiment 2's comparison of experiential information at time of order/purchase (taste-before in Experiment 2) conditions and no experiential information at the time of order/purchase (taste-after in Experiment 2) conditions. In the experience-before condition, the three pens were placed in the front inner pocket of the binder containing the catalog page in the expectation that participants would remove, examine, and even try these pens when asked to look at the catalog page. However, this manipulation proved ineffective. The experimenter observed that the participants did not attempt to examine the pens physically. It may have been the case that the pens were pushed too deeply into the binder pocket or that the instructions, in an attempt to maintain similar language between conditions, did not explicitly encourage trial. Thus, both conditions effectively represented the experience-after, or no experiential information before order, condition. Differences between the catalog and bricks-and-mortar conditions were not significant for any of the dependent variables, and therefore the results were collapsed across this factor.

Method

Subjects. Participants were undergraduates from a large university who received class credit for participation in two sessions held two days apart. All subjects attended both sessions. Subjects participated in groups of three to nine. Data were collected from 84 subjects. Data from two subjects were eliminated from analysis because of a failure to follow instructions and a large number of missing responses. Among the remaining 82 subjects, 59 ordered products and 23 did not order products.

Design. A 2 (lenient versus restricted policy) × 2 (experience-before versus experience-after) between-subjects design was used. However, as previously mentioned, the manipulation of the experience-before versus experience-after conditions was ineffective, and these cells of the design were collapsed for data analysis. Subjects were randomly assigned to conditions. Key dependent variables were amount of continued search, order and return rates, deliberation time for the order decision, and deliberation time for the keep-or-return decision.

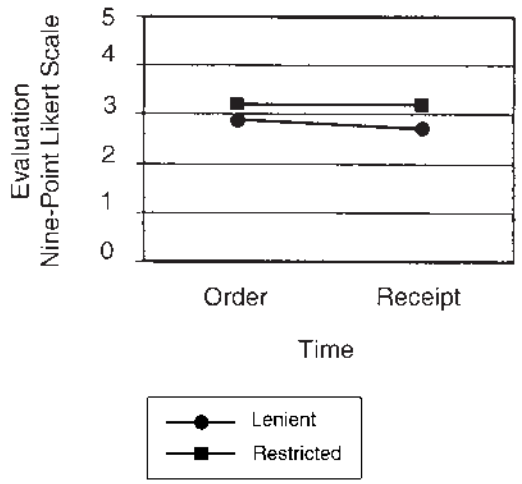
Procedure. The general procedure of the study was similar to that of Experiments 1 and 2. Participants were presented with a concept-testing study for a proposed catalog of business accessories from an unnamed manufacturer. The study consisted of two 30-minute sessions conducted two days apart. Participants were given the opportunity to order a catalog product at Session 1 and later received the product, and they were given an opportunity to keep or return that product at Session 2.

At Session 1, participants were situated at a carrel containing a study booklet, a closed hardcover binder, a closed manila folder, and a digital clock showing the hour, minute, and second. Participants were orally instructed to follow all instructions in the booklet and work at their own pace. The first page of the booklet explained the catalog cover story. The second page explained the presence of the digital clock in their carrel. Participants were told that this study, though a "real and important" study in its own right, was also to be used to help refine a future, large-scale study. To provide the researchers with information that would assist in timing and planning the future study, subjects would be periodically asked to record the time, as displayed on the digital clock (including seconds), in their study booklet. In truth, this exercise provided time measures of the decision to order, the decision to keep or return, and the amount of continued search.

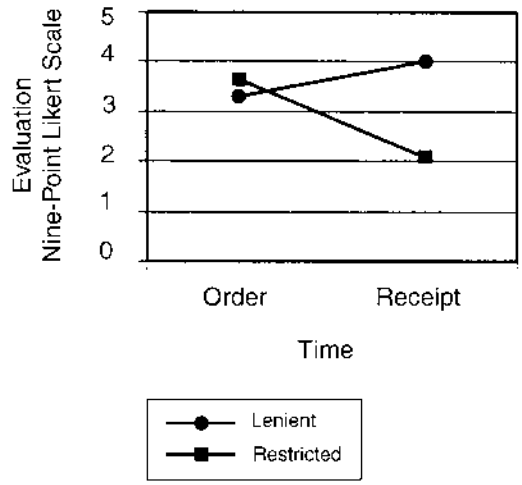
Participants next read and evaluated several of the catalog's policies, including the return policy. As previously stated, return policy leniency was manipulated through the inclusion (or not) of the original shipping and handling charge and return postage costs in the refund (in this case, a value of $1.50). Thus, subjects initially chose either a pen whose value equaled $3.00 plus the $1.50 shipping and handling charge or a gift certificate to a local store for $4.50. Subjects who ordered a pen later faced one of two conditions: (1) In the lenient policy condition, they could return the pen for a gift certificate for the full $4.50, or (2) in the restricted policy condition, they could return the pen for a $3.00 gift certificate ($1.50 subtracted for nonrefund of shipping and handling charges). Participants then were

Figure 4
DIFFERENCES DUE TO TIMING OF PRODUCT EXPERIENCE

A: Quality Evaluations in Experience Before Conditions



B: Quality Evaluations in Experience After Conditions



asked to open the binder located in their carrel and examine a mock catalog page, which featured three of the pens offered by the manufacturer. Participants then completed a filler questionnaire regarding the catalog page.

Next, participants were offered the opportunity to order a pen or, in exchange, take a gift certificate for similar merchandise at a nearby store. Participants were instructed to keep their binders open during this decision. The time measures on this page recorded the decision to order deliberation time.

The next task involved the opportunity to continue the product search by looking at information about competitor's pens. Participants were asked to open the manila folder that contained several pages of information about similarly priced pens. The instructions explained that in a real-world catalog purchase, a consumer would have the ability to continue looking for a product during the time the ordered product was being delivered. Therefore, they were told to look at the information in the manila folder as much or as little as they wanted. The time measure for this task provided a behavioral measure of amount of continued search. This concluded Session 1.

At Session 2, participants returned to their original carrels. Present in each carrel was the same digital clock, a Session 2 study booklet, and the binder that now contained both the catalog page and the ordered pen packaged in bubble wrap. Participants who did not order a pen received a gift certificate to a university bookstore. Participants were again orally instructed to proceed through the booklet at their own pace.

The first task involved product receipt. Participants were instructed to open their binders and examine and sample their ordered pens. Participants were reminded that they could return the pen for the gift certificate and were asked to indicate this keep-or-return decision. The time recorded for this task is the keep-or-return decision deliberation time.

Results

Continued search. Table 3 describes the results for all measures. The behavioral measure of continued search also supports the hypothesis that a lenient return policy can inhibit continued search for a product during the period between order and receipt. Consistent with the pilot study results and H_3, the amount of time subjects spent looking at information on competitor's pens was significantly lower in the lenient policy condition (m = 119.25 seconds) than in the restricted condition (m = 158.13; t = 2.19, $p = .03$).

Order and return rates. The percentage of orders and the percentage of returns, though not formally tested, appears to differ between lenient and restricted policy conditions in the expected pattern. The percentage of orders in the lenient policy condition was 81.8% but only 60.5% in the restricted policy condition ($\chi^2 = 3.766$, $p < .05$, two-tailed); this corresponds with the results of Experiment 2. The return rate, however, was 13.9% in the lenient policy condition and 4.3% in the restricted policy condition ($\chi^2 = 1.857$, $p < .173$, two-tailed). Although this type of result will necessarily differ by product category and may not be generalizable, it is interesting to note that more people ordered and kept the pen in the lenient policy condition than in the restricted policy condition. However, the implications of these order/return rate results should be tempered by the recognition of the limitations of experimental research in addressing these issues.

Replication of order and keep-or-return decision times. This study also sought to replicate the decision time results of Studies 1 and 2. As in the preceding studies, the decision to order took significantly longer for those in the restricted policy condition (m = 141.2 seconds) than those in the lenient policy condition (m = 123.8 seconds; t(55) = 1.63, $p < .05$, one-tailed). Also as predicted, the keep-or-return decision times did not differ significantly between the restricted policy condition (m = 90.6 seconds) and the lenient policy condition (m = 85.8 seconds; t(55) = .57, $p < .29$, one-tailed).[4] The pattern of reduced deliberation in

[4]The average deliberation time for subjects who decided not to order was 123 seconds in the lenient policy condition and 108 seconds in the restricted policy condition.

Table 3
EXPERIMENT 3 AND PILOT STUDY 3 RESULTS

	Experiment 3 Results				
	Continued Search Time	*Order Deliberation Time*	*Keep-or-Return Deliberation Time*	*Order Rate*	*Return Rate*
Lenient	119.25 seconds	123.8 seconds	85.8 seconds	81.8%	13.9%
Restricted	158.13 seconds	141.2 seconds	90.6	60.5%	4.3%
t	2.19	1.63	.57	$\chi^2 = 3.766$	$\chi^2 = 1.857$
p-Value	.03	.05	.29	.05	.173

	Pilot Study 3 Results	
	Likelihood to Order	*Likelihood to Continue Search*
Lenient	60.7%	39.3%
Restricted	46.6%	68%
F/t	8.84	176.9
p-Value	.01	.001

lenient policy conditions (H_1) is again supported. Furthermore, it might be argued that return times were influenced by more people returning products in the lenient policy condition than in the restricted policy condition. However, a return decision took longer on average than a keep decision, which should increase the average return time in the lenient policy condition. The influence of this increased time (due to a greater number of returns) is counter to the direction of our predictions and thus creates a more conservative test of the hypothesis and stronger support.

GENERAL DISCUSSION

This research finds consistent effects of return policy on remote purchase behavior. Although some effects seem obvious, especially those pertaining to the first decision stage (i.e., order), the existence of a distinct second decision stage leads to several more complex and nonintuitive effects of return policy leniency. Greater lenience in return policy resulted in an increased probability of order. Choice probabilities of remote purchase were affected by return policy, even when leniency was manipulated in different ways. Lenient return policies decreased the probability of continued product search during the time between order and delayed receipt. Lenient return policies reduced deliberation time for the decision to order but did not result in a subsequent increase in deliberation for the keep-or-return decision; thus, lenience appears to decrease the overall amount of purchase decision conflict. Greater lenience in return policies also resulted in more positive ratings of product quality, both before choice and after receipt. Delay affected deliberation time. Subjects spent less time over the decision to order in lenient conditions only when there was a delay between order and receipt. With a delay, subjects' attitudes about product quality became more positive in lenient conditions and more negative in restricted conditions. Finally, the data suggest that these results are unique to remote purchases; no such effects were observed in the conditions that simulated in-store purchase characteristics.

It is interesting to note how the results depart from a dissonance-based explanation. Predictions derived from cognitive dissonance theory would run counter to those predicted by quality signaling. With cognitive dissonance, subjects ordering under a restricted policy should evaluate products more favorably, because their ability to reverse the choice decision is more difficult or costly. Without these restrictions, subjects in the lenient condition should not be motivated to increase product evaluations to reduce dissonance. Thus, the overall effect of return policy observed in this study is more consistent with a signaling explanation.

Furthermore, note the conceptual similarity of remote purchases to bricks-and-mortar purchases in which a consumer makes a choice without sufficient evaluation of experiential attributes (e.g., a consumer buys a set of flannel bedsheets that is wrapped in plastic). Thus, some of the effects identified in this research may also apply to certain more traditional purchases as well. However, it should be noted that there is greater variance in return policy leniency among remote retailers than among bricks-and-mortar retailers, and more homogeneous expectations of leniency may reduce the strength of potential signaling effects.

Theoretical Implications

The theoretical contributions of this work are threefold. First, this article characterizes the most fundamental differences between traditional bricks-and-mortar channels and emerging forms (electronic or otherwise) of remote purchase. Consumer use of e-commerce can be a challenging topic to study because of the difficulty in predicting e-commerce capabilities and constructing models of these capabilities in the lab. Rather than focus on futuristic retailing formats that are difficult for both consumers to imagine and researchers to create, this study examines the characteristics of remote purchase through analogous catalog formats.

Second, the results of these studies also lend support to the concept of endowment, or mere ownership, effects (see Beggan 1992; Kahneman, Knetsch, and Thaler 1990). Specifically, the increase in purchase commitment prior to product receipt observed in these studies is similar to Sen and Johnson's (1997) findings. Their research demonstrates that consumers can exhibit mere ownership effects (i.e., an increase in product evaluation due to possession) before actual ownership or consumption. Taken together, these results suggest that product ownership depends more on cognitive perception than physical possession.

Third, this work contributes to the signaling literature in two ways. First, it provides empirical evidence of quality

inferences based on return policy. Many articles on product warranties as economic signals are analytical models (three show empirical evidence), and no papers to date show empirical evidence for return policies, a similar but distinct risk minimization tool (Kirmani and Rao 2000). Second, the results of this research suggest that contrary to economic models of signaling, sellers can cheat by sending false signals. Because of endowment effects in remote purchase, consumers do not appear to correct signal-inflated expectations of quality after product receipt.

Managerial Implications

Lenient return policies appear to offer several benefits that may be contrary to managerial intuition and practice. Greater lenience appears to increase order rates by reducing the amount of conflict in the decision to order and creating higher quality expectations. This much managers may expect. However, managers may fear that these effects will have consequences that could lead to higher return rates after the product is received. Specifically, decision conflict (as measured by deliberation time) may increase at the time of the keep-or-return decision, and inflated expectations may lead to lower quality perceptions after examination. The results of these studies suggest that these fears may be unwarranted. Keep-or-return decision deliberation did not increase in lenient conditions and was not significantly greater than in restricted conditions, which suggests that the conflict associated with this decision did not similarly increase. Quality perceptions were higher in lenient conditions than in restricted conditions even after physical examination. Even with the increased flexibility that lenient return policies allow, subjects showed greater commitment to their order by being less likely to continue product search for competing alternatives.

Finally, the results of this study suggest that the positive effects of lenient return policies are unique to remote vendors. Benefits demonstrated in catalog conditions did not hold for bricks-and-mortar conditions in which products could be tried before order. These differences do not appear to be due to the separation of the purchase decision into the decision to order and the keep-or-return decision but rather are due to the delay between these two decisions and the inability to examine goods directly in remote venues.

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